

Experiment 09: Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger

AIM: Creation of Triggers, and perform insert, update, delete triggers.

Definition: Triggers are stored programs, which are automatically executed or fired when some events occur.

Triggers are occurred when we perform any of the following events:

- 1.A database manipulation (DML) statements (DELETE, INSERT, UPDATE).
- 2.A database definition (DDL) statements (CREATE, ALTER, And DROP).
- 3.A database operation (SERVERERROR, LOGON, LOGOFF, STARTUP, or SHUTDOWN).

Triggers could be defined on the table, view, schema, or database with which the event is associated.

Benefits of Triggers:

Triggers can be written for the following purposes:

1. Generating some derived column values automatically.
2. Enforcing referential integrity.
3. Event logging and storing information on table access.
4. Auditing.
5. Synchronous replication of tables.
6. Imposing security authorizations.
7. Preventing invalid transactions.

Syntax:

Create or replace trigger <trigger name> (before/after/instead of) {insert/delete/update} on <table name> [for each statement/row] [when <condition>]

Declare

<Declarations>

Begin

<Executable statements>

End;

SQL> set serverOutput on;

Example:

Write a PL/SQL code for creation of trigger to insert data into a table.

```
SQL>create or replace trigger t1
before
insert on sailors
for each row
begin
:new.sname:=upper(:new.sname);
end;
/ Trigger created
```

Output:

```
insert into sailors values(22,'dUstin');
select * from sailors;
```

SID	SNAME
----	-----
22	DUSTIN

Write a PL/SQL code for creation of trigger to update data into a table.

```
SQL>create trigger t2
after update of sid on sailors
for each row
begin
if(:new.sid<80) then
raise_application_error(-20017,'cant update');
end if;
```

end;

/

Trigger created

Output:

SQL> update sailors set sid=79 where rating=10;

update sailors set sid=95 where rating=10;

*

ERROR at line 1:

ORA-20018: you cant update this row

ORA-06512: at "PRASANNA.TRG3", line 3

ORA-04088: error during execution of trigger

'PRASANNA.TRG3'

Write a PL/SQL code for creation of trigger to delete data from a table.

SQL>create trigger t6

after

delete on sailors

for each row

begin

if :old.sid=22 then

raise_application_error

(-20019,'you can't delete this row');

end if;

end;

/

Trigger created

Output:

```
SQL> delete from sailors where sid=22;
```

```
delete from sailors where sid=22
```

```
      *
```

```
ERROR at line 1:
```

```
ORA-20010: you cant delete this row
```

```
ORA-06512: at "PRASANNA.TRG2", line 3
```

```
ORA-04088: error during execution of trigger
```

```
'PRASANNA.TRG2'
```